

The Multiplicative Lattice of Gamma Values at Rational Arguments:

Complete Reduction, Trig-Root Phenomena, and Machine-Checkable Multiplier Certificates

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Abstract

In logarithmic coordinates, every consequence of the reflection and Gauss multiplication formulas for Γ at rational arguments is a $\mathbb{Q}$ -linear relation, and by the Koblitz–Ogus theorem these relations generate *all* multiplicative relations among such values (modulo algebraic numbers and powers of π). One exact row reduction per denominator grid therefore replaces open-ended saturation search: it yields the complete reduction table, the grid’s irreducible basis, and a provable termination criterion at once. We record the resulting reducer (Wolfram and Python implementations with identical output; the 550-rule legacy database of a saturation prover is covered 550/550, and all 151 grids it spanned regenerate, complete and verified, in seconds), its trigamma analogue (3428 rules over grids 3–100, basis dimension $\varphi(N)/2$), and an explanation with census of the “trig-root” Γ identities: D -th roots of sine products arise from eliminations passing through a target with multiplicity D , with degrees up to 16 observed, and the root degree is a property of the derivation path, not of the value. The same row-combination matrices are *multiplier certificates*: machine-checkable derivations, replayed by an independent checker with no search and no numerics in the certification path. We describe the certificate model and its axiom layer, and four findings the linear algebra produced: Schwarz conjugation is a necessary axiom (a measured rank deficiency of exactly one dimension); within the stored law vocabulary Catalan’s constant is definitional, and weight-2 dilogarithm values on the imaginary line need exactly that one new constant; prime denominators require the full multiplication row, which is precisely where irrational coefficients enter; and a trusted constant layer must be explicitly invoked, not assumed. The program’s ledger at the repository snapshot: 212 derived + 4 definitional certificates replaying green, including 82/83 real-argument order-2 reduction rules of a 437-rule legacy table, with the single holdout and the Γ -side frontier documented.

1 Introduction

A saturation prover for Γ -value identities — enumerate instances of the Gauss multiplication theorem, normalize, rewrite with the accumulated rule base, record whatever survives — slows down in a characteristic way: it keeps rediscovering known identities, and genuinely new ones

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arrive at ever-longer intervals. The analysis of one such prover (`GammaProver.nb`, a Wolfram notebook from a 2014–2018 prover campaign, with a recovered database of 550 rules and 1117 examined arguments, denominators up to 308) located the cause structurally: span-membership of a candidate is discovered only by paying the full normalize-substitute-simplify price; a two-minute `FullSimplify` budget sits on the hot path; and the enumeration has no notion of completeness, so it keeps spending candidates on exhausted grids.

The decisive observation is that the problem is linear algebra in disguise — and that, unusually, the linear-algebra reformulation is not merely faster but *complete*, by a classical theorem. This article records the reformulation and everything it bought: benchmarks, a complete census of root phenomena, the trigamma analogue, and the conversion of the machinery into machine-checkable proof certificates for the project’s identity stores.

Epistemic status. The lattice constructions and certificates are exact (rational linear algebra over symbolic right-hand sides); their mathematical completeness rests on the Koblitz–Ogus theorem, cited, not reproved. Every emitted reduction is additionally verified numerically (40 digits in the Wolfram implementation, 50 in Python, 10^{-30} tolerance for the legacy-database checks). Certificates are replayed by an independent checker in which the certification path is purely exact (no numerics); numerics appear only in two proposal/guard roles described in Section 6.

2 The lattice model

Work on the denominator- N grid, in coordinates $x_k = \log \Gamma(k/N)$, $k = 1, \dots, N-1$. The two classical law families become exact $\mathbb{Q}$ -linear rows with symbolic right-hand sides over the atoms $\log \pi$, $\log n$, $\log \sin(\pi k/N)$:

- reflection, for $k < N/2$:

$$x_k + x_{N-k} = \log \pi - \log \sin\left(\frac{\pi k}{N}\right);$$

- Gauss multiplication, for each divisor $d \geq 2$ of N and $k = 1, \dots, N/d$:

$$\sum_{j=0}^{d-1} x_{k+jN/d} - [dk < N] x_{dk} = \frac{d-1}{2} (\log 2 + \log \pi) + \left(\frac{1}{2} - \frac{dk}{N}\right) \log d,$$

the instance at $z = k/N$ of the Gauss multiplication theorem [2],

$$\prod_{j=0}^{d-1} \Gamma\left(z + \frac{j}{d}\right) = (2\pi)^{(d-1)/2} d^{1/2-dz} \Gamma(dz),$$

with the convention $x_N := \log \Gamma(1) = 0$ in the boundary case $dk = N$.

The matrix is small ($O(N)$ rows $\times$ $(N-1)$ columns). One exact row reduction of the augmented block $[M \mid I]$ over $\mathbb{Q}$ (columns ordered numerator-descending, so pivots land on large numerators) yields simultaneously:

1. a reduction $\Gamma(k/N) = \exp(\text{elementary}) \cdot \prod \Gamma(\text{basis})^{\mathbb{Q}}$ for *every* non-basis k at once, the elementary side computed exactly as (combination matrix) $\cdot$ (right-hand-side vector) — no `FullSimplify` anywhere;

2. the grid’s *basis* — its irreducible Γ -values — and with it a termination guarantee: the lattice rank says exactly how many independent values remain.

Law 2.1 (completeness; Koblitz–Ogus [1]). *The reflection and multiplication relations generate all multiplicative relations among Γ -values at rational arguments, modulo algebraic numbers and powers of π . Relative to this classical result the lattice output is the complete answer that the saturation loop was approximating from below.*

Result 2.2 (benchmarks and coverage). *Implementations: `tools/gamma-reducer.wl` (Wolfram) and `tools/gamma_reducer.py` (pure Python: exact $\mathbb{Q}$ arithmetic via `fractions.Fraction`, sparse atom-coefficient right-hand sides, `mpmath` verification). Both produce identical output (same rules, same bases).*

<i>workload (generation only)</i>	<i>Wolfram</i>	<i>Python</i>
<i>denominator-120 grid (103 rules, 16-element basis)</i>	0.42 s	0.056 s
<i>all 151 legacy-database grids, denominators ≤ 308 (15 588 rules)</i>	75.3 s	7.44 s
<i>mpmath verification of all 15 588 rules</i>	—	8.32 s

Coverage of the 550 recovered `GammaRules.wdx` arguments: 550/550. The original prover needed multi-hour sessions for 550 rules with no terminus; either implementation beats it by orders of magnitude with a completeness guarantee. Sample output (the sine values collapse to radicals on their own):

$$\Gamma\left(\frac{5}{12}\right) = \frac{2^{2/3} \sqrt{\pi} \Gamma\left(\frac{1}{12}\right)}{(1 + \sqrt{3}) \Gamma\left(\frac{1}{6}\right)}.$$

Remark 2.3. The row-combination matrix produced by the augmented reduction is, at the same time, a *multiplier certificate* in the sense of Section 6: the reducer’s output is born proved, not merely verified.

3 Trig-root Γ identities: mechanism and census

The prompt for this investigation was the identity

$$\frac{\Gamma\left(\frac{2}{7}\right)\Gamma\left(\frac{11}{42}\right)}{\Gamma\left(\frac{1}{21}\right)} = \frac{8 \sin\left(\frac{\pi}{7}\right) \sqrt{\pi \sin\left(\frac{\pi}{21}\right) \sin\left(\frac{4\pi}{21}\right) \sin\left(\frac{5\pi}{21}\right)}}{2^{1/42} 7^{1/3} 3^{9/28}},$$

of the kind that has surprised mathematicians before the mechanism was recognized [verified RootReduce-exactly against the legacy table].

Result 3.1 (the mechanism, in lattice terms). *When the lattice solve expresses a target $\log \Gamma(k/N)$, the elimination may pass through a combination in which the target appears with multiplicity $D > 1$ — i.e. the relation natively proves $\Gamma(k/N)^D = (\text{elementary}) \cdot (\Gamma\text{-monomial})$. Dividing by D puts coefficient $1/D$ on the elementary $\log \sin$ atoms, and exponentiating yields a D -th root of a sine product; positivity of all factors fixes the branch. The root is a property of the derivation path, not of the value: the same $\Gamma(11/42)$ carries a square root in the harvested spelling above and cube roots in the lattice’s own reduction — both exactly correct.*

The surprise dissolves further at grids whose sines are radical-expressible: for $N = 30, 60$ the renderer collapses $\sqrt{\sin \cdots}$ into golden-ratio radicals and nothing looks trig-rooted at all. The phenomenon is only *visible* when the sine arguments have large totient $(21, 39, 78, \dots)$, where no real-radical form exists (casus irreducibilis).

Result 3.2 (census, denominator grids $N \leq 200$, primitive arguments). 856 *lattice reductions* carry fractional $\log \sin$ coefficients. By root degree D (the lcm of the coefficient denominators):

D	2	3	4	6	8	16
rules	616	46	106	26	42	20

Demanding integral Γ -exponents (so only the trigonometric part is rooted, as in the prompt identity) leaves exactly 10 rules, all $D = 4$, at $\Gamma(71/78)$, $\Gamma(85/156)$, $\Gamma(65/114)$, $\Gamma(167/198)$ and conjugate-numerator siblings. The $D = 16$ cases live on the $N = 168$ grid with 26 sines under the radical.

Result 3.3 (showpieces). [verified to 30+ digits in both implementations] The fourth-root champion ($N = 78$; $\varphi(78) = 24$, so none of the cosines collapse to radicals):

$$\Gamma\left(\frac{71}{78}\right) = 2 \cdot 2^{55/312} \pi \sqrt{\cos \frac{\pi}{78}} \cos^{1/8} \frac{\pi}{39} \cos^{3/4} \frac{\pi}{26} \cos^{5/8} \frac{2\pi}{39} \cos^{11/8} \frac{4\pi}{39} \left(\cos \frac{\pi}{13} \sec \frac{7\pi}{78}\right)^{3/8} \Gamma\left(\frac{1}{39}\right) \\ \bigg/ \left[13^{1/48} \cos^{1/4} \frac{5\pi}{78} \cos^{3/2} \frac{3\pi}{26} \cos^{7/8} \frac{5\pi}{39} \cos^{5/8} \frac{11\pi}{78} \Gamma\left(\frac{2}{39}\right) \Gamma\left(\frac{5}{78}\right) \right],$$

and the lattice's cube-root reduction of the prompt's own $\Gamma(11/42)$:

$$\Gamma\left(\frac{11}{42}\right) = \frac{7^{1/6} \sqrt{\pi} \Gamma\left(\frac{1}{42}\right) \Gamma\left(\frac{1}{21}\right) \left(\sec \frac{\pi}{42} \sec \frac{\pi}{21}\right)^{2/3} \sec^2 \frac{\pi}{14} \left(\cos \frac{2\pi}{21} \sec \frac{5\pi}{42}\right)^{1/3}}{8 \cdot 2^{5/21} 3^{3/7} \Gamma\left(\frac{2}{21}\right) \Gamma\left(\frac{3}{14}\right)}.$$

4 The trigamma analogue

The same construction applies one weight up. On the grid $y_k = \psi^{(1)}(k/N)$ the lattice rows are the reflection law $\psi^{(1)}(k/N) + \psi^{(1)}(1 - k/N) = \pi^2 \csc^2(\pi k/N)$ and the multiplication theorem $\psi^{(1)}(dk/N) = d^{-2} \sum_{j=0}^{d-1} \psi^{(1)}(k/N + j/d)$, with the pin $\psi^{(1)}(1) = \pi^2/6$ wherever an argument hits 1; the elementary atoms are π^2 and the $\pi^2 \csc^2(\pi k/N)$ values.

Result 4.1. *tools/psi_reducer.py* derives the complete trigamma reduction tables for grids $N = 3, \dots, 100$: 3428 rules, all *mpmath*-verified, in 7.7 s total, with basis dimensions equal to $\varphi(N)/2$ throughout — the same dimension count as the cyclotomic-polylogarithm lattice, which the exact DFT bridge between the two worlds explains (the two relation lattices are Fourier-conjugate).

This subsumes, by a factor of ~ 50 , a legacy 70-entry hand/PSLQ-era workbook of $\psi^{(1)}(k/n)$ closed forms (denominators 2–16), which is preserved separately as a harvested store.

5 The harvest: legacy notebooks, verified and taxonomized

A kernel-free pipeline (notebook box parser, custom `TraditionalForm/InputForm` interpreter, *mpmath* 60-digit verification) mined the owner's OneDrive notebook corpus into verified identity stores:

- `harvested-gammaprover-rules.wl` — the recovered 550-rule saturation-prover database, every rule verified in-kernel to $|N[\Gamma(\arg) - \text{closed form}, 40]| < 10^{-30}$, zero failures;
- `harvested-gamma-values.wl` — 508 verified Γ reductions from four further notebook families (including a 451-rule table recovered from a single 700 KB output cell, and 637 sine-product reductions from a working-table pair identified during the sweep);
- `harvested-trigamma-values.wl` — all 70 identities of the trigamma workbook, mpmath-verified at 60 digits and re-verified in-kernel.

Result 5.1 (novelty taxonomy). *Measured against the 670-rule **FunctionTransformations** Γ table [4] (denominators ≤ 120) that anchors the corpus: 314 distinct harvested arguments lie beyond the table — 280 ceiling-breakers from the prover database (denominators 123–308) plus 34 arguments that only the notebook harvest reaches: one further ceiling-breaker, $\Gamma(67/160)$, and 33 upper-half reflection twins of table arguments. (Eight more harvest ceiling-breakers duplicate prover-database arguments; an earlier recorded decomposition “280 + 8 + 34” double-counted them.)*

6 Multiplier certificates

Before this program, every identity in the project’s stores carried one badge: *numerically verified* ($\gtrsim 25$ digits). The certificate program adds a qualitatively stronger badge: *derived* — a recorded, machine-checkable certificate exhibiting the value as an exact linear combination of verified functional-equation instances, replayed with no search and no numerics in the certification path by an independent checker running under the project’s one-command regression guard.

6.1 The model

A certificate stores law *names* plus concrete substitution strings (never expressions — a corrupted certificate cannot smuggle in a false instance) and two exact multiplier vectors over a canonical real/imaginary splitting of the instances. Validity means the claim lhs – rhs equals the recorded linear combination *identically*: every inert Li/ψ atom coefficient cancels **RootReduce**-exactly, and the elementary residual is killed by a dedicated sound zero test (certified principal-branch log normalization, certified multiplicative log-basis reduction, formal coefficient comparison). Numerics appear in exactly two proposal/guard roles: proposing root-of-unity decompositions that exact algebra then certifies, and a per-instance branch-domain guard (an instance violating its law’s branch domain is generically numerically false and is rejected). Linear systems subsume orbit words, fixed-point arguments, and multi-value systems in one format.

A *derived* certificate proves its claim *relative to* a declared axiom layer: (a) the classical laws of the project’s functional-equation store [3] — Euler reflection, inversion, Landen, duplication/triplication/multiplication, the five-term family, the Kummer-class relations, Schwarz conjugation, the cyclotomic polygamma sum, and the polygamma reflection/multiplication laws, each numerically verified at recorded test points; (b) definitional pins ($\text{Li}_n(1) = \zeta(n)$, $\psi^{(m)}(1) = (-1)^{m+1}m!\zeta(m+1)$, and the two Catalan pins below); (c) Euler’s $\zeta(2k) \in \mathbb{Q}\pi^{2k}$ and trigonometric values at rational multiples of π ; (d) **RootReduce** as the exact algebraic-number oracle.

6.2 Findings the linear algebra produced

Finding 6.1 (Schwarz conjugation is a necessary axiom). *The anharmonic orbit of i is the six-point family $\{i, -i, 1 \pm i, (1 \pm i)/2\}$, and inversion and duplication at i produce the same sum relation. The full system of reflection/inversion/Landen/duplication instances over the orbit is rank-deficient by exactly one dimension, and the missing direction is precisely $\text{Li}_2(\bar{z}) = \overline{\text{Li}_2(z)}$. No combination of the single-variable transformation laws separates $\text{Li}_2(i)$ from $\text{Li}_2(-i)$ — the transformation laws are Möbius/algebraic in the argument and blind to conjugation. Obvious in hindsight; the linear algebra measured it.*

Finding 6.2 (Catalan is definitional; the rest is not). $\beta(2)$ admits no derivation from the stored laws: it enters every certificate that needs it through one of two recorded definitional pins, both termwise series identities,

$$\text{Im Li}_2(i) = G \quad \text{or} \quad \psi^{(1)}\left(\frac{1}{4}\right) - \psi^{(1)}\left(\frac{3}{4}\right) = 16 G.$$

By contrast $\text{Li}_2(-1) = -\pi^2/12$ (duplication at $z = 1$), the four golden-ratio dilogarithm values, $\text{Li}_3(\frac{1}{2})$, the golden trilogarithm $\text{Li}_3(1/\varphi^2)$, and $\text{Li}_2(2)$ are all honestly derived; the flagship compaction is $\text{Li}_2(\frac{1}{2})$ as a one-instance certificate (Euler reflection at its own fixed point), and each golden-ratio dilogarithm carries exactly the classical four-equation linear system, the coincidence $1 - 1/\varphi = 1/\varphi^2 = (1/\varphi)^2$ doing all the work. Within this law vocabulary, the weight-2 dilogarithm world has exactly one new constant on the imaginary line, and it is Catalan.

Finding 6.3 (prime denominators need the full multiplication row). *For the cyclotomic values $\text{Li}_n(e^{2\pi i p/q})$ the polygamma vocabulary pins the ψ -grid via proper-divisor multiplication instances when q is composite; for prime q the only ψ -sum relation is the full row $d = q$ — the multiplication theorem at $z = 1/q$,*

$$\psi^{(m)}(1) = q^{-m-1} \sum_{k=1}^q \psi^{(m)}\left(\frac{k}{q}\right),$$

whose $k = q$ term is $\psi^{(m)}(1)$ itself — and it is exactly the source of the irrational coefficients — e.g. the $\sqrt{5}\zeta(3)$ terms in the $q = 5$ trilogarithm closed forms. Omitting $d = q$ cost precisely the prime- q certificates and nothing else.

Finding 6.4 (a trusted layer must be applied, not assumed). *Wolfram auto-evaluates $\sin(\pi/4)$ but not $\sin(\pi/8)$: the $q = 8$ certificates initially failed because $\csc^2(\pi/8)$ survived as an opaque atom in the formal zero test while being numerically zero all along. The fix is policy, not code volume: trig at rational multiples of π is part of the declared trusted base and is applied explicitly during law instantiation and residual checking. A trusted constant layer is only trusted if the checker actually invokes it; silent reliance on auto-evaluation makes the trust boundary version-dependent.*

6.3 Coverage ledger

At the repository snapshot, the certificate ledger replayed by the independent checker under the regression guard is 212 **derived** + 4 **definitional**, **zero failures**:

store	derived / attempted	notes
curated values	45 + 4 definitional	every concrete curated value
legacy polylog table, order-2 real	82/83	one honest holdout (below)
legacy Γ table, denominators ≤ 40	85/100	complete-lattice law pool

The Γ certificates use the complete denominator- q relation lattice of Section 2 as the law pool (20/20 on the pilot slice, versus 3/20 for the earlier heuristic pool). The polylog order-2 sweep was driven by three cumulative generator upgrades — atom-closure five-term pairs with structured one-parameter pair families $(x, 1 - x)$, $(x, x/(x - 1))$, $(x, -x)$; square-root argument seeding for duplication ladders; and a height-weighted cluster retry that pools orbit vocabulary across related rules without ever using a related *claim* (no circularity) — the largest certificate being 80 instances.

Negative result 6.5 (the honest holdout). $\text{Li}_2(-7/10)$, *the deepest reduction ladder in its slice (10 residual dilogarithms), remains rank-deficient even at deep search caps: the orbit pairs demand an internal 17-flavoured cancellation that no implemented pair family reaches. The natural next lever is second-order stray matching or ladder relations beyond the stored law set. On the Γ side, the frontier is 15 no-certs of $q = 40$ grade (sine-product exact algebra blow-ups) and the $q > 40$ tail, awaiting cyclotomic-unit arithmetic or certificate transport through the exact DFT bridge to the lighter ψ -side atoms.*

7 Open directions

1. Emit the full mega-store of grid reductions (e.g. all $N \leq 360$) superseding the legacy database’s coverage, with the row-combination matrices landing directly as replayable certificates.
2. Certificate transport between the cyclotomic-polylogarithm world and the ψ -grid world, through the exact $\mathbb{Q}(\zeta_q)$ DFT matrix.
3. The 33 unproved exotic PSLQ-discovered values (the Im Li_3 family at $\mathbb{Q}(i)/\mathbb{Q}(\sqrt{3})$ points), expected to need the two-variable five-term law instanced over algebraic pairs.
4. Single-valued (Bloch–Wigner) certificates, removing the branch-domain numeric guard for the real-analytic subset of claims; and formalization of the small trusted replay core (the checker’s linear algebra plus the elementary zero test) in a proof assistant.

Sources and verification artifacts

This article presents results recorded in the Smithereens repository (`src/PolyLog`) in the commit range 272af472..68236dab4 (2026-06-09 through 2026-06-11): the lattice reducer and its analysis in `docs/reports/gammaprover-analysis-and-lattice-reducer__38c270ec5048.md`; the trig-root mechanism and census in `docs/reports/trigroot-gamma-identities__6f71bf0491ba.md`; the certificate model in `docs/proof-certificates.md` and the round reports `docs/reports/phase3-multiplier-certificates-round1__6822403edd2f.md`; the stores under `identities/`; the tools under `tools/` (`gamma-reducer.wl`, `gamma_reducer.py`, `psi_reducer.py`, `proof-core.wl`, `proof-kit.wl`, `check-proofs.wl`).

References

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 - [3] L. Lewin, *Polylogarithms and Associated Functions*, North-Holland, 1981. (The dilogarithm functional equations: reflection, inversion, Landen, duplication, the five-term relation.)
 - [4] V. Reshetnikov, *FunctionTransformations* (Wolfram Language package), 2016, <https://github.com/VladimirReshetnikov/FunctionTransformations>. (The 437-rule polylog and 670-rule Γ reduction tables anchoring the corpus.)